The Uniquantsystem: Integer Time Counting on a 1/23 Scale with a Defined Anchor and Distributed Residuals

Joern Christian Anders, B.Sc. Process Technology

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Abstract

We describe the Uniquantsystem, a time count that uses only integers and stays attached to the Sun without accumulating error. The unit is the quantum q = 1/23 tick = 75/1012 s; a cycle has 66 ticks plus a zero crossing, a day 768 cycles, a year 13 × 28 days plus a zero day, a chart 23 years and a great chart 23^2 = 529 years. Counting starts at the calibration point 00, named jj: Friday, 13 July 2029, 00:00 CEST; until then the parameters are calibrated against observation, afterwards they are defined. Year boundaries follow a defined anchor, a parabola with integer coefficients over the denominator 529, fitted to the apparent solar crossings of the longitude the Sun occupies at 00. All residuals are absorbed in the cycle zero crossings, which last 1 or 2 q, so every cycle boundary lies within q/2 = 0.037 s of the anchor and any run of units deviates by less than 1 q. We call this property quantum irrelevance and prove it. The residual state repeats exactly every 529 years, and the residuals at chart starts run through all 23 classes k/23 q. The true Sun deviates from the anchor periodically by −17.7 to +16.9 min; eight terms of known origin describe this to 0.71 min. A second axis counts sidereal time for ascendant and house cusps and agrees with the Swiss Ephemeris to 3.1″ over 2029–2264. A third axis counts the lunar phase in joe, 1/1 165 824 of a lunation, and follows the smoothed lunar phases to 3.2 s. System time runs one Jules, 69.159 s, ahead of UTC. A fourth axis classifies the position of Earth's pole in fields of 1 Iris = 1″/299 and forecasts it as a probability per field. Further axes measure ΔT in Jules, count the solar and lunar terms as integer epicycles (residuals 0.71 and 3.2 min) and derive eclipses and tides from the lunar and nodal axes. A convergence study answers Plato's question of how many uniform circles the motions require: 24 circles bring the Sun to 1″ and 109 the Moon, Mars needs 113 for 1′, and two Kepler ellipses with 22 parameters match 140 circles. Fitted to 100 years and tested on the next 100, ellipses plus circles forecast the Sun to 1.8″ and Mars to 19″, about eight times better than circles alone. A reference implementation verifies all bounds in exact rational arithmetic.

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Keywords: calendar arithmetic · cumulative rounding · error feedback · sigma-delta quantisation · Sturmian sequences · time scales · sidereal time · lunar phase · continued fractions · polar motion · eclipses · tides · epicycles · frequency analysis

1 Introduction

Calendars must reconcile periods that are not integer multiples of each other: the second, the day and the tropical year. Leap rules do this with rational approximations. The fractional day of the vernal-equinox year, 0.2423768 d, has the continued fraction [0; 4, 7, 1, 18, …] with the convergents 1/4, 7/29, 8/33 and 151/623, which drift by 11 min, 86 s, 4.1 s and 0.1 s per year. Every fixed rule drifts linearly, because the true ratio is irrational and varies slowly.

This work takes a different route. Each integer quantity is obtained by rounding its exact cumulative target, $P_n = \lfloor x_n + ½ \rfloor$, so the error stays below half a unit for all n. Computed incrementally, this is a first-order error-feedback (sigma-delta) quantiser behind exact integrators: the residual is carried as state instead of being discarded.

Three decisions complete the design. First, the target is defined rather than modelled: an integer-coefficient parabola, calibrated once against observation at a calibration point 00. Second, all residuals are absorbed at the lowest level, in the zero crossings of the cycles, so years and charts need no correction terms of their own. Third, nature enters only as a measured deviation from the definition. Sections 2 to 11 develop the construction, Section 12 asks how many uniform circles the motions require and how well they forecast, Section 13 collects the error budget and Section 14 the verification.

2 Structure of the count

All lengths are exact fractions because the tick is tied to the SI second: 768 × 66 ticks = 86 400 s, hence 1 tick = 75/44 s and q = 75/1012 s ≈ 0.0741107 s.

Unit	Composition	Length in q	Length in s
Quantum q	1/23 tick	1	0.0741107
Tick	23 q	23	1.7045455
Cycle	66 ticks + zero crossing of 1 q (2 q with an extra quantum)	1 519 or 1 520	112.5741 or 112.6482
Day	768 cycles, 3 or 4 of them with an extra quantum	1 166 595 or 1 166 596	86 457.139 or 86 457.213

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Unit	Composition	Length in q	Length in s
Month	28 days	≈ 32 664 678	≈ 2 420 801
Year	13 months + zero day = 280 320 cycles, J of them with an extra quantum	425 806 080 + J, J = 1 290 … 1 331	31 556 774.704 + 0.0741 J
Chart	23 years	9 793 539 840 + ΣJ (chart 0: 9 793 570 437)	≈ 725 808 086
Great chart	23 charts = 529 years	residual exactly 0 at its end	≈ 1.67×10^{10}
Jules (Ju)	offset of system time from UTC; unit of the rotation lag ΔT	933.18544 = 5 832 409/6 250	69.159
joe	1/1 165 824 lunation = 1.11166″ of mean elongation	29 or 30, mean 29.530589	mean 2.18853
Lunation	768 lunar cycles × 66 lunar ticks × 23 joe	34 427 469 + 0 … 4	≈ 2 551 443

The nominal lengths of cycle, day and year are coprime to the grid 23 (phases 1, 9 and 19 mod 23), so no residual can hide behind a common boundary; two scales m_1 and m_2 share $\log_2 \gcd(m_1, m_2)$ bits of information about a position. The system day is a counting unit, not a solar day. On average it is 57.2 s longer than 86 400 s and loses one day against the civil calendar every 4.14 years.

3 Calibration point jj, system time and system longitude

Definition 1 (calibration point and system time). 00, named jj, is Friday, 13 July 2029, 00:00:00 CEST = 12 July 2029, 22:00:00 UTC; the start day jj is chart 0, year 0, month 0, day 0. System time is UTC + 1 Ju, with one Jules 1 Ju = 69.159 s = 5 832 409/6 250 q ≈ 933.18544 q; it runs 25 ms behind TT = UTC + 69.184 s, and JD_TT(jj) = 2462330.417467. Before jj the parameters are calibrated against observation; from jj on they are fixed, and every deviation of nature is recorded as a series that starts at zero.

From 20 May 2027 UTC runs without leap seconds (28th CGPM, Resolution 5). TAI − UTC therefore stays 37 s and both offsets are permanent, while UT1 − UTC may grow to 3 600 s. Counts in q measure durations since jj, so the fractional 0.18544 q of the Jules never enters them. Physically, 69.159 s is a value of ΔT = TT − UT1, for UT1 − UTC = +25 ms; ΔT is the accumulated lag of Earth rotation, and its secular growth comes mainly from lunar tidal friction.

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Definition 2 (system longitude). $\lambda_sys = \lambda - \lambda_0$, where λ is the apparent geocentric ecliptic longitude of the Sun referred to the true equinox of date and $\lambda_0 = \lambda(00) = 110.82764°$ (provisional, VSOP87). At 00 the Sun is at $\lambda_sys = 0°$ by construction. Because λ_0 is a constant in tropical longitude, the origin of system longitude moves with the equinox and the seasons stay fixed.

The apparent longitude is $\lambda = \lambda_geo + \Delta\psi - 20.4898''/R$, with the geometric longitude λ_geo from VSOP87, the nutation in longitude $\Delta\psi$ and the annual aberration at solar distance R in au. At the true equinoxes given by PyEphem this expression returns 0° to within 0.25″. All coordinates (chart, year, month, day, cycle, tick, quantum) are zero-based and equal 0 at jj. The vernal equinox remains the origin of tropical longitude but is no longer a year boundary.

4 The defined anchor

Definition 3 (anchor). With Y = 425 806 080 q and x the time since 00 in years,

$\tau(x) = Y{\cdot}x + (b{\cdot}x + g{\cdot}x^2) / d$, with b = 703 993, g = −11 and $d = 23^2 = 529$.

The year boundaries are $A_n = \mathrm{round}(\tau(n))$ and the residuals $\rho_n = \tau(n) - A_n \in [-½, ½)$.

Derivation. For n = 0 … 966 the instants t_n at which the true Sun reaches λ_0 were computed by Newton iteration on the apparent longitude (tolerance 10^{-10} degrees). The excesses $e_n = t_n - Y{\cdot}n$ were decomposed by least squares into a cubic trend and eight periodic terms (Section 7). The trend was shifted by its value at n = 0 so that the anchor passes through 00. Integers b and g were then chosen to minimise $\max_n |(b{\cdot}n + g{\cdot}n^2)/d - \mathrm{trend}_n|$; for fixed g this objective is convex in b, so an integer ternary search is exact. The anchor deviates from the smoothed course by at most 57.5 s over 2029–2995 and by at most 14 s until 2052.

Theorem 1 (period 529). $\rho_\{n+529\} = \rho_n$ for all n, and $\rho_\{529k\} = 0$ for every integer k.

Proof. $\tau(n + 529) - \tau(n) = 529{\cdot}Y + b + g{\cdot}(2n + 529)$ is an integer, so the fractional part of τ is unchanged; τ(0) = 0 gives $\rho_0 = 0$. No shorter period exists: a period p requires $2g{\cdot}p/529 \in \mathbb{Z}$, and $2g = -22$ is coprime to 23. ∎

Theorem 2 (residue classes). The residuals at chart starts, $\rho_\{23j\}$ for j = 0 … 22, take each value k/23 with k = −11 … 11 exactly once.

Proof. $\tau(23j) = 23j{\cdot}Y + b{\cdot}j/23 + g{\cdot}j^2$, hence $\rho_\{23j\} \equiv b{\cdot}j/23 \pmod 1$. Since $b \equiv 9 \pmod{23}$ is coprime to 23, $j \mapsto 9j \bmod 23$ is a permutation. ∎

The chart-start residuals therefore advance in steps of 9/23 q and average exactly to zero over every great chart: the 1/23 scale reappears one level below the quantum.

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Annual jump. $J(n) = A_{n+1} - A_n - Y$ lies in $\{\lfloor \Delta_n \rfloor, \lceil \Delta_n \rceil\}$ with $\Delta_n = (b + g\cdot(2n + 1))/d$. In the first year $\Delta_0 = 1\,330.78$ q and $J = 1\,331$ q = 98.6 s; by 2995 J falls to 1 290 q, and $J(n + 529) - J(n) = 2g = -22$ q exactly. Being a quadratic, τ can be advanced with two additions per year, as in a difference engine.

5 Distributed counting

All residuals are absorbed in the zero crossings of the cycles. With $Z = 365 \cdot 768 = 280\,320$ cycles per year, the boundary of cycle c, counted from 00, is

$B(c) = \mathrm{round}(\tau(c / Z))$.

Proposition 1. $B(c + 1) - B(c) \in \{1\,519, 1\,520\}$ for every c in the horizon, and the number of 1 520-q cycles in year n equals $J(n)$.

Proof. τ increases by $1\,519 + J/Z \approx 1\,519.0047$ q per cycle, and two roundings of values that differ by δ differ by $\lfloor \delta \rfloor$ or $\lceil \delta \rceil$. A year contains exactly Z cycles, so its surplus over $Z \cdot 1\,519 = Y$ is $B(Z\cdot(n + 1)) - B(Z\cdot n) - Y = J(n)$. ∎

The extra quanta form a Sturmian sequence: their spacing takes only two adjacent values.

Year	Extra quanta	Spacing in cycles	Per day
2029	1 331	210 or 211 (≈ 6.6 h)	3 or 4
2512	1 311	213 or 214	3 or 4
2994	1 290	217 or 218	3 or 4

Bound. $|B(c) - \tau(c/Z)| \leqslant q/2$ for every c. Year and chart boundaries are ordinary cycle boundaries and inherit the bound; neither a year insertion nor a boundary element is needed.

Coordinates. For a time t, the cycle index is the largest c with $B(c) \leqslant t$, found from the inverse of the parabola and an exact correction of a few steps. Year $n = \lfloor c/Z \rfloor$, chart $\lfloor n/23 \rfloor$, day and cycle follow from c; the offset $o = t - B(c)$ gives tick and quantum for $o < 1\,518$, otherwise the zero crossing.

Countdown. Zero crossings are labelled backwards: −1 for a 1-q crossing, −2 and −1 for a 2-q crossing. Every zero is thus preceded by −1, no fold flag is required, and the map between time and coordinates is bijective. For example, t = 0 is chart 0, year 0, month 0, day 0, cycle 0, tick 0, quantum 0; t = 1 518 is the zero crossing of cycle 0; t = 425 807 411 starts year 1.

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6 Quantum irrelevance

Definition 4. A residual is quantum-irrelevant, with respect to the counting quantum q, if it (i) is at most q/2 at every boundary, (ii) contributes less than 1 q over any window, (iii) vanishes on average and (iv) is exactly computable from the anchor, so that it need neither be stored nor displayed. The term refers to the quantum q of the count, not to quantum physics.

Theorem 3. All rounding residuals of the distributed count are quantum-irrelevant.

Proof. (i) follows from B(c) = round(τ(c/Z)). (ii) The units from a to b have total length B(b) − B(a) = τ_b − τ_a − (ρ_b − ρ_a); the error is a difference of two numbers in [−½, ½) and therefore smaller than 1 (telescoping). (iii) The chart-start residuals average exactly to zero over every great chart by Theorem 2; at year starts the mean over 966 years is +0.0009 q. (iv) ρ = τ − round(τ) is a function of the anchor. ∎

Verification. At the 967 year starts up to 2995 the largest residual is 0.499 q, the largest window error 0.996 q, the mean +0.0009 q and the RMS 0.284 q ≈ 1/√12, the value of a uniform distribution.

Method	Largest total error over 966 years	Growth
Each year rounded independently, constant year length	214 q = 15.8 s	linear
Stochastic rounding, median of 200 runs	13.5 q = 1.0 s	∝ √N
Cumulative rounding with carried residual	⩽ 0.5 q = 0.037 s	bounded

The deviations of nature are not quantum-irrelevant: they oscillate by minutes or grow, and must be described or measured (Sections 7 and 8).

7 Nature minus definition

The true Sun reaches λ_0 up to 17.7 min earlier or 16.9 min later than the anchor. Around the cubic trend the true crossings scatter by 6.59 min; eight periodic terms of known origin reduce the scatter to 0.71 min, with a largest residual of 2.0 min.

Source	Harmonic	Apparent period	Amplitude
Lunar node (nutation)	1	18.61 yr	7.22 min
Venus, synodic	2	3.98 yr	3.25 min

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Source	Harmonic	Apparent period	Amplitude
Jupiter, synodic	1	11.86 yr	3.05 min
Lunar synodic month	1	2.72 yr	2.67 min
Venus, synodic	1	2.67 yr	1.99 min
Mars, synodic	2	15.77 yr	1.48 min
Venus, synodic	3	8.10 yr	0.62 min
Jupiter, synodic	2	5.93 yr	0.54 min

Aliasing. The series is sampled once per year, so a frequency f in cycles per year appears at f − round(f). This explains the apparent periods.

Calibration freezes the phase. At 00 the true Sun was 0.52 min behind its smoothed course (periodic part −0.05 min, unexplained residual +34 s). Because the anchor passes through 00, this offset remains as the mean of nature minus definition, −0.49 min over 2029–2995. It belongs to the variance observed up to 00.

Comparison with the equinox. For the true vernal equinox against the Meeus mean-equinox polynomial, the same decomposition gives ±19 min and reduces the scatter from 6.8 to 0.66 min. The sources are the same; the amplitudes depend on the position in the orbit.

8 The rotation axis

A second axis counts Earth rotation for ascendant and house cusps. Both axes start at jj: the annual axis at the Sun on λ_sys = 0°, the rotation axis at the equinox on the meridian, that is, at sidereal time.

Motion of Earth's axis	Form	Size	In the system
Rotation	about the instantaneous axis	Earth rotation angle, strictly linear in UT1	counted on the grid
Precession (cone)	axis circles the ecliptic pole; half-angle = obliquity 23.44°	≈ 25 800 yr; 50.3″ per year in longitude, 0.32° per chart	polynomial (IAU 2006)
Nutation	waves on the cone	main term 18.6 yr; 17.2″ in longitude, 9.2″ in obliquity	periodic terms

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Motion of Earth's axis	Form	Size	In the system
Polar motion	axis moves within the crust	fractions of an arcsecond	not modelled

Formulae.

- Earth rotation angle: ERA = 2π · (0.7790572732640 + 1.00273781191135448 · D_U), with D_U = JD(UT1) − 2451545.0
- Mean sidereal time: GMST = ERA + p(T), p = 0.014506″ + 4612.156534″·T + 1.3915817″·T² − …, T in Julian centuries of TT from J2000
- Apparent sidereal time: GAST = GMST + Δψ·cos ε
- Mean obliquity: ε_0 = 84381.406″ − 46.836769″·T − 0.0001831″·T² + …
- MC = atan2(sin RAMC, cos RAMC·cos ε); ASC = atan2(cos RAMC, −(sin RAMC·cos ε + tan φ·sin ε))

Stellar versus equinox reference. A sidereal day referred to the stars (ERA) lasts 86 164.0989 s, referred to the equinox (GMST) 86 164.0905 s, which is 8.4 ms less. This difference produces the growing offset between the tropical and the sidereal zodiac.

Grid. One rotation has 768 × 66 × 23 = 1 165 824 rotation quanta; one rotation quantum is 1.11″ or 0.074 s of sidereal time. The accumulated sidereal time is rounded and the residual kept as state.

Defined rotation. The coupling to system time is ΔT = TT − UT1. From 00 on, ΔT is defined as a polynomial, ΔT_def = 69.2 s + 0.28 s/yr · (J − 2027) + 0.0032 s/yr² · (J − 2027)², provisionally 69.93 s at 00, 77.3 s in 2050, 106.7 s in 2100 and 384 s in 2300. Value, rate and curvature are fixed at 00 from IERS data. Later measurements form the series ΔT − ΔT_def; each second of difference shifts the true meridian by −15.04″ relative to the defined one, and one Jules by −17.34′. In Jules, ΔT_def is 1.011 Ju at jj, 1.118 Ju in 2050, 1.543 Ju in 2100 and 5.55 Ju in 2300.

UT1 from UTC. Without leap seconds UT1 − UTC grows. Civil input therefore uses TT = UTC + 69.184 s and UT1 from the measured UT1 − UTC or from ΔT_def: UT1 − UTC = 69.184 s − ΔT_def, provisionally −0.75 s at jj, −8.2 s in 2050 and −37.5 s in 2100.

House cusps. Placidus, Porphyry, equal and whole-sign houses are implemented; Placidus is undefined beyond the polar circles. Against the Swiss Ephemeris, ascendant, MC and cusps agree to 3.1″ over 2029–2264; later the precession models diverge (28″ in 2497, 101″ in 2732). At 00 the ascendant in Berlin is at 11°09.4′ Aries and the MC at 4°06.7′ Capricorn, which are 260.33° and 163.28° in system longitude. An uncertainty of ±20 s in ΔT shifts the ascendant by about ±5′.

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9 The lunar axis

A third axis counts the lunar phase on the grid of the day and of the rotation axis: one lunation is 768 lunar cycles × 66 lunar ticks × 23 joe = 1 165 824 = $2^9 \cdot 3^2 \cdot 11 \cdot 23$ joe. One joe is 1/1 165 824 of the mean elongation, 1.11166″, and lasts on average 29.530589 q = 2.18853 s; the ratio joe : q equals the length of the synodic month in days.

Definition 5 (lunar axis). The mean phase at jj is rounded to whole joe, Φ_0 = 65 331, that is 20.17° after mean new moon. The instant of the phase $\Phi_0 + 1\,165\,824 \cdot x$ is

$\nu(x) = (A \cdot x + G \cdot x^2) / 23^4$ q, with A = 9 634 217 446 503, G = 32 and x the time since jj in lunations.

Joe m, counted from jj, starts at round(ν(m / 1 165 824)); lunation k starts at m = 1 165 824·k − Φ_0. Phase 0 is the mean new moon, not jj, so first quarter, full moon and last quarter (291 456, 582 912 and 874 368 joe) are integers. Lunation 0 began at the mean new moon of 11 July 2029, 06:17:01 UTC; jj itself is lunation 0, lunar cycle 43, lunar tick 2, joe 11.

Derivation. The reference is the mean lunar phase of Meeus (1998, eq. 49.1), JDE = $2451550.09766 + 29.530588861 \cdot k + 0.00015437 \cdot T^2 - 0.000000150 \cdot T^3 + 0.00000000073 \cdot T^4$ with T = k/1236.85; at jj, k = 365.056039. Shifted through jj, the parabola was fitted by integer minimax over 11 950 lunations. It deviates from the smoothed course by at most 3.2 s, and the offset frozen at jj is −0.91 s.

Proposition 2. Every joe lasts 29 or 30 q, every lunation 34 427 469 + J_M q with J_M ∈ {0, …, 4}, and every lunar boundary lies within q/2 of ν, so the lunar residuals are quantum-irrelevant (Definition 4).

Proof. As for Proposition 1 and Theorem 3, with ν in place of τ: ν grows by 29.5306 q per joe, two roundings of values δ apart differ by ⌊δ⌋ or ⌈δ⌉, and window errors telescope. Over the 11 949 new-moon boundaries up to 2995 the largest window error is 0.9998 q. ∎

Nature minus definition. True new moons (PyEphem, apparent conjunction) lie between −14.11 and +14.06 h from the defined ones over 2029–2995, with RMS 7.50 h and mean −0.30 min; in lunation 0 the true new moon came 9.57 h later. Eight terms of known origin reduce the scatter to 3.2 min (largest residual 14.5 min).

Term	Origin	Apparent period	Amplitude
M′	lunar anomaly, elliptic lunar orbit	411.78 d	9.77 h
M	solar anomaly, elliptic Earth orbit	365.26 d	4.08 h
2M′	second harmonic of the lunar anomaly	205.89 d	23.1 min
2F	distance from the node	173.31 d	15.0 min

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Term	Origin	Apparent period	Amplitude
M′ − M	apsidal cycle	3 232.86 d = 8.85 yr	10.5 min
M′ + M	combination	193.56 d	7.3 min
2M	second harmonic of the solar anomaly	182.63 d	2.9 min
3M′	third harmonic of the lunar anomaly	137.26 d	0.6 min

Sampling once per lunation folds the anomalistic month onto 411.78 d, just as annual sampling folds the synodic month onto 2.72 yr on the annual axis.

Lunations per year. With year starts taken from the anchor, a system year contains 12 defined new moons (611 years) or 13 (355 years), and every chart has 8 or 9 years with 13. At jj a system year holds 12.3682460 lunations = [12; 2, 1, 2, 1, 1, 15, 4, 26]. The convergents 99/8, 136/11, 235/19 (Meton) and 3 661/296 miss by +38.3, −35.9, +2.4 and −0.6 h. The seventh partial quotient is 15 rather than 17 because the year at λ_0, 365.24159 d, is shorter than the mean tropical year.

Eclipse cycles. The saros (223 lunations = 18.030 system years) matches the node argument to −0.48° and the anomaly to −2.8°. The inex (358) matches the node to +0.04° but not the anomaly (−117.5°), and the Metonic cycle misses the node by 7.57°. In joe the saros is exactly 223 · 1 165 824.

10 The polar axis

A fourth axis classifies the position of Earth's pole on the crust. One Iris is 1 Ir = 1″/299 with 299 = 13 · 23, that is 3.3445 mas or 10.33 cm on the ground (mean Earth radius 6 371 km). The field of a pole with IERS coordinates x, y in arcseconds is (round(299·x), round(299·y)), a Gaussian integer X + iY.

Definition 6 (polar axis). The pole is not counted but measured. Its class is its Iris field, and its forecast is a probability distribution over the fields.

Polar motion obeys the Liouville equation $m + (i/\sigma_c)\cdot\dot{m} = \chi$ with $m = x - i\cdot y$ and $\sigma_c = (2\pi/T_c)(1 + i/2Q)$, $T_c \approx 433$ d. The excitation χ by atmosphere, ocean and hydrology is random, so the pole cannot be defined in advance.

Path 1962–2026. In IERS 20 C04 (23 648 daily values at 0 h UTC) the pole lies between x = −92 and 97 Ir and y = −7 and 178 Ir, and the median daily step is 0.72 Ir. On 29 September 2026 it was in field (53, 97).

Forecast model. The mean pole is fitted linearly over 10 years together with the circles. Annual and semiannual ellipses (each two counter-rotating circles) and Chandler circles (433 d) are then refitted over the last 3 years, and the residual at the start decays with 120

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days. In hindcasts this beats a pure 3-year model at 1 000 days (complex RMS 19.5 versus 22.4 Ir).

Hindcast 2010–2026. Forecasts start every 15 days; for each horizon the table gives the error spread in x and y, the radii holding 68 % and 95 % of the probability, and the probability that the most likely field is hit.

Horizon	σ_x (Ir)	σ_y (Ir)	68 % radius (Ir)	95 % radius (Ir)	P(field)	Cases
1 day	0.20	0.16	0.3	0.4	98.7 %	407
7 days	1.14	0.99	1.6	2.6	13.1 %	407
30 days	3.70	3.56	5.5	8.9	1.2 %	406
90 days	7.21	7.34	11.0	17.8	0.3 %	402
180 days	8.83	8.96	13.4	21.8	0.2 %	396
1 year	9.74	9.80	14.8	23.9	0.2 %	383
1 000 days	13.61	14.02	20.9	33.8	0.1 %	341

Probability per field. With a normal distribution around the forecast (μ_x, μ_y),

$$P(X, Y) = [\Phi((X + ½ - \mu_x)/\sigma_x) - \Phi((X - ½ - \mu_x)/\sigma_x)] \cdot [\Phi((Y + ½ - \mu_y)/\sigma_y) - \Phi((Y - ½ - \mu_y)/\sigma_y)],$$

and the probabilities of all fields sum to 1.

Forecast for jj. From 29 September 2026, 1 018 days ahead, the centre is (69.8, 162.6) Ir. The most likely field (70, 163) has 0.08 %; 68 % of the probability lies within 21 Ir (2.2 m) and 95 % within 34 Ir (3.5 m).

Complex state space. The fields form the lattice $\mathbb{Z}[i]$, and the state of the polar axis is a probability distribution p over it. A forecast acts as a stochastic matrix ($p \mapsto P \cdot p$) and every measurement as a Bayesian update. This is classical probability, not quantum mechanics: there is no interference. In the Koopman–von Neumann picture the periodic circles correspond to eigenfunctions with eigenvalues $e^{i\omega t}$; the random excitation forms the remainder that only measurements constrain.

11 Further axes: ΔT, epicycles, eclipses and tides

ΔT in Jules. ΔT = TT − UT1 = 32.184 s + (TAI − UTC) − (UT1 − UTC) is formed daily from 1 January 1972 (IERS 20 C04 and the leap-second table, 19 996 days), counted in fields of 1 q and expressed in Jules. On 29 September 2026 it was 69.2046 s = 1.00066 Ju (field 934 q), 1.3 ms below ΔT_def. ΔT passed 1 Ju = 69.159 s on 26 October 2018, 31 July 2024 and 21 May 2026. A forecast from the last measured value with slope and annual, semiannual, monthly

and fortnightly terms of the last year has hindcast errors of 0.3 ms after 1 day, 7.2 ms after 30 days, 83 ms (1.1 q) after one year and 0.36 s (4.9 q) after 1 000 days. For jj it predicts 69.52 ± 0.36 s = 1.0052 Ju; the provisional ΔT_def = 69.93 s lies 1.1 σ above.

Epicycle axes. Each periodic term of nature minus definition is a uniform circular motion. Its phase is counted on the grid L = 1 165 824 per turn by cumulative rounding, $p(n) = \mathrm{round}(L\cdot(\varphi_0 + f\cdot n))$, and contributes $A\cdot\cos(2\pi\cdot p(n)/L)$. With the eight solar terms the scatter of the true crossings of λ_0 falls from 6.59 to 0.71 min (largest 2.0 min); with the eight lunar terms the scatter of the true new moons falls from 7.5 h to 3.2 min (largest 14.5 min). The counted circles reproduce the floating-point decomposition, because half a grid step of phase changes a contribution by at most $\pi/L \approx 2.7 \times 10^{-6}$ of its amplitude.

Nodal axis and eclipses. The Moon's argument of latitude F (Meeus 1998, eq. 49) advances by 30.67° per lunation and is counted on the same grid. At the true syzygies (PyEphem) a solar eclipse occurs if $|\beta| < \pi_M - \pi_S + s_M + s_S$, and a lunar eclipse in the penumbra if $|\beta| < 1.02\cdot(\pi_M + \pi_S + s_S) + s_M$, with lunar latitude β, parallaxes π and apparent radii s. Over 2029–2229 the node distance of the mean syzygy separates certain from impossible eclipses.

Kind	Number	Per year	Certain below	Impossible above
Solar eclipse	465	2.33	15.12°	19.62°
of which central	296	1.48	8.03°	13.71°
Lunar eclipse (penumbral)	470	2.35	14.44°	20.38°
of which umbral	299	1.50	8.71°	13.69°

Lunation 0 begins with the solar eclipse of 11 July 2029; the next are 5 December 2029 (solar), 20 December 2029 (lunar, umbral) and 1 June 2030 (solar, central).

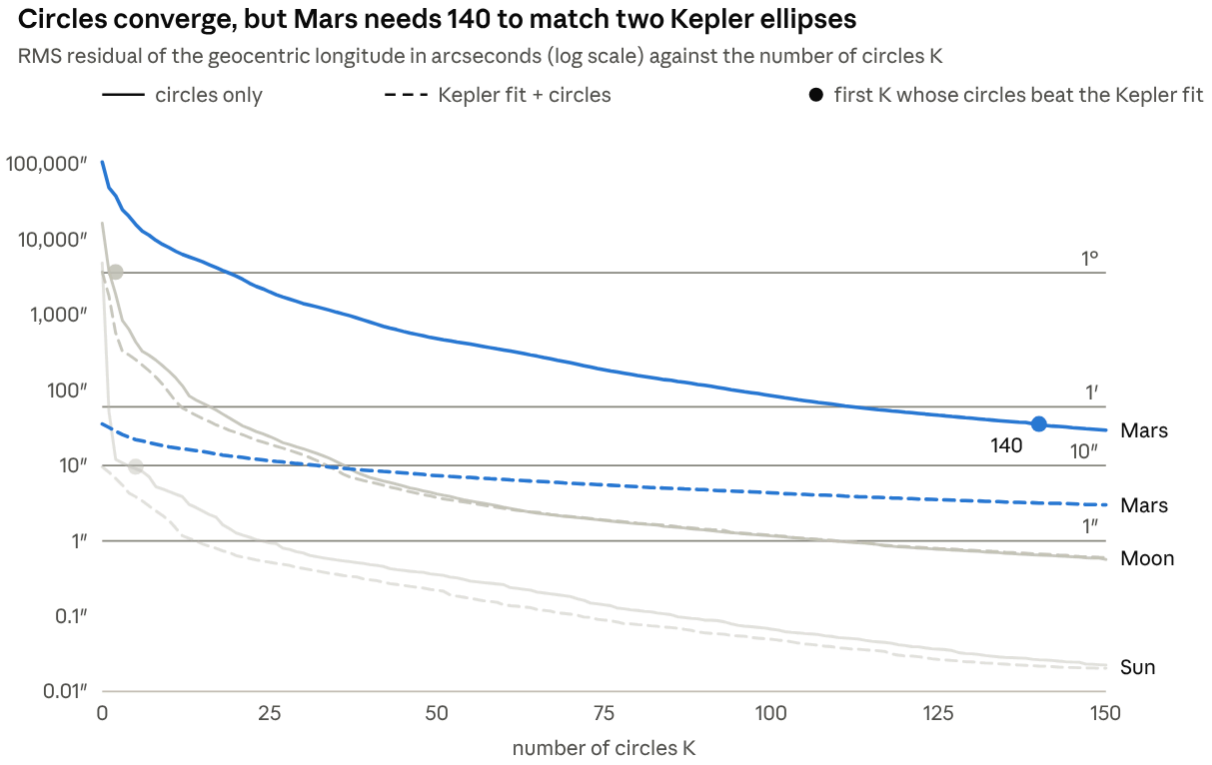
Tides. S2, half a pure day, lasts exactly 582 912 q = L/2 q, and the spring–neap cycle, half a lunation, exactly 582 912 joe = L/2 joe (14.7653 d). The S2 phase is twice the solar hour angle and the spring–neap phase twice the elongation in joe, so M2 = S2 − spring–neap and $f_{M2} = (2/L)\cdot(1/q - 1/joe)$: the principal lunar tide (12.420601 h) is the doubled beat between the q and joe grids. Local tidal lags come from measurements.

12 Plato's question: how many circles?

Plato set astronomy the task of saving the phenomena with uniform circular motions alone. The task can always be solved: a quasi-periodic motion is an almost periodic function in the sense of Bohr (1925) and can be approximated uniformly by finite sums of uniform circular motions, and epicycles on a deferent can trace practically any path

(Hanson 1960). What remains open is the price, that is, how many circles a given accuracy requires. We measure it for three bodies from jj on: the geocentric longitudes of the Sun and of Mars over 200 years and of the Moon over 100 years, from PyEphem, geometric and referred to the mean ecliptic and equinox of date, at 36 525 epochs each.

Method. The longitude is represented as a quadratic mean motion plus K circles, each with its own frequency, amplitude and phase, so that K circles cost 3 + 3K parameters. The frequencies are found one at a time and without prior knowledge from the spectrum of the residual (Hann window, fourfold zero padding, golden-section refinement of the peak); after each step all circles are refitted together by least squares, implemented as continued Gram–Schmidt orthogonalisation. This is frequency analysis in the sense of Laskar (2003). For comparison, Kepler ellipses with linearly varying elements (11 parameters each, starting values from Meeus 1998, ch. 31 and 47) are fitted to the heliocentric positions of Earth and Mars and to the geocentric positions of the Moon, and the same search for circles is then run on the residual of the ellipses.



zyklus23.py platon · PyEphem 4.2.1, geocentric longitude from jj: Sun and Mars 200 years, Moon 100 years, 36 525 epochs each

Circles per accuracy. The table gives the number of circles for an RMS residual of 1°, 1′, 10″ and 1″, and the smallest K at which the circles alone beat the Kepler fit.

Body	1°	1′	10″	1″	Kepler fit	Circles that beat it
Sun	1	1	4	24	1 ellipse, 11 parameters: 9.6″	5
Moon	2	17	37	109	1 ellipse, 11 parameters: 1.03°	2

Body	1°	1′	10″	1″	Kepler fit	Circles that beat it
Mars	19	113	over 150	over 150	2 ellipses, 22 parameters: 35.5″	140

The first circles are the historical ones. For the Sun the first circle is the anomalistic year with an amplitude of 1.91°, the eccentre of Hipparchus; it leaves 52″. For the Moon the search returns, in this order, the anomaly (6.29°, Hipparchus), the evection (1.27°, Ptolemy), the variation (0.66°, Tycho Brahe), the second harmonic of the anomaly (0.21°) and the annual equation (0.18°, Tycho Brahe and Kepler). For Mars the first circle is the synodic period of 779.9 d with 37.2°, Ptolemy's epicycle that produces the retrograde loop, and the third is the sidereal period of 687.0 d with 11.3°, the eccentre.

Convergence, but not economy. The residual falls with every circle, so Plato's programme converges for all three bodies; only its cost differs. The Sun's orbit is nearly circular (e = 0.017), and 24 circles reach 1″. The Sun perturbs the Moon and spreads its motion over many lines, so the Moon needs 109 circles for 1″. The geocentric loop of Mars combines two eccentric orbits: 113 circles reach 1′, while two Kepler ellipses with 22 parameters reach 35.5″, which takes 140 circles with 423 parameters. The right curve thus replaces more than a hundred circles; this is the step from Ptolemy to Kepler. For the Moon the ellipse is worth only two circles because the Sun shapes its orbit: with or without the ellipse, about 110 circles are needed for 1″, the price of the three-body problem.

In the system. Every circle found is a uniform phase and can be counted as an epicycle axis on the grid L (Section 11), with a rounding residual below half a grid step. The convergence curve thus states how many integer counters an axis needs for a given accuracy.

Forecast beyond the window. Tracing a window is not yet saving the phenomena; a model should also continue them. All models are therefore fitted to the first 100 years from jj and measured on the following 100 years, the Sun and Mars every 2 d and the Moon daily. For the Sun and Mars, ellipse plus circles carries only a constant offset, because the mean motion and its change are already part of the ellipse; for the Moon a quadratic term is added, the secular acceleration that the ellipse lacks.

Body	Circles, fit (K = 150)	Circles, best forecast	Kepler fit, fit → forecast	Kepler fit + circles, best forecast
Sun	0.02″	13.4″ (K = 10)	9.6″ → 9.7″	1.8″ (K = 14)
Moon	0.57″	8.7″ (K = 150)	1.03° → 1.03°	10.6″ (K = 150)
Mars	32″	156″ (K = 128)	34.6″ → 39.2″	19.0″ (K = 65)

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Circles describe, ellipses explain. Inside the window the residual of the circles keeps falling; outside it stalls, for the Sun at 13 to 17″ while the fit reaches 0.02″. Additional circles reproduce slow changes such as the drift of eccentricity and perihelion, through beats of neighbouring periods (357 to 371 d for the Sun) and through periods as long as the window (up to 95 years). Beyond the window these beats do not continue the drift. The Kepler ellipse carries its slowly varying elements across the boundary: for the Sun its residual is the same inside and outside (9.6″ and 9.7″), for Mars it grows only from 34.6″ to 39.2″. Ellipse plus circles forecasts best, 1.8″ for the Sun and 19″ for Mars, about eight times better than circles alone; modern planetary theories are built the same way, from orbital elements and periodic perturbations. For the Moon the ellipse again adds nothing. Its best forecast keeps a quadratic term, the secular acceleration found by Halley in 1695; without it, ellipse plus circles falls back to 33″. Over the first decade after the window all models stay close to their fit (the Moon with 150 circles at 1.6″, the Sun with 30 circles at 3.6″): circles suffice for decades, centuries need the law.

13 Error budget

All grid errors stay below q/2 = 0.037 s and do not accumulate. The periodic deviation of the true Sun is about 29 000 times larger than the largest of them.

Effect	Size	Treatment
Cycle boundary against anchor	⩽ q/2 = 0.037 s	cumulative rounding
Any run of units	< 1 q = 0.074 s	telescoping
Anchor against smoothed course of the true Sun	⩽ 57.5 s (⩽ 14 s until 2052)	definition at 00
Offset frozen at 00	−0.52 min	part of the observed variance
True Sun against anchor	−17.7 to +16.9 min; scatter 6.6 min, 0.71 min after eight terms	described, not counted
Lunar axis against smoothed lunar phases (Meeus 49.1)	⩽ 3.2 s; offset frozen at jj −0.91 s	definition at jj
True new moon against lunar axis	−14.1 to +14.1 h; RMS 7.5 h, 3.2 min after eight terms	described, not counted

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Effect	Size	Treatment
Pole forecast (Iris)	68 %: 0.3 Ir after 1 day, 5.5 Ir after 30 days, 14.8 Ir after 1 year	measured, forecast as probability per field
ΔT forecast (Jules)	σ = 7 ms after 30 days, 83 ms after 1 year, 0.36 s after 1 000 days	measured, forecast as probability per field (1 q)
True Sun and true Moon with epicycle axes	0.71 min (Sun), 3.2 min (Moon), RMS	counted, residual described
Ephemeris VSOP87 (provisional)	≈ 1″, i.e. ≈ 24 s	replaced by DE440 at 00
ΔT of the rotation axis	±20 s ≈ ±5′ in the ascendant	measured series ΔT − ΔT_def
Cosmic expansion in the bound solar system	≈ 5×10^{-14} s per 23 years	negligible

14 Implementation and verification

The reference implementation of the Uniquantsystem, zyklus23.py (Python, exact rational arithmetic with fractions.Fraction) checks all bounds at start-up; a full run with reports takes about 15 s, and python3 zyklus23.py platon runs the convergence study of Section 12 in about 1.5 min and python3 zyklus23.py vorhersage the forecast test in just over 1 min. The anchor derivation, the periodic decomposition and the convergence study need PyEphem and NumPy, the Kepler fits also SciPy, the comparison of house cusps the Swiss Ephemeris (pyswisseph). The checks cover:

- Calibration point jj: 12 July 2029, 22:00:00 UTC = 13 July, 00:00 CEST, a Friday; JD_TT = 2462330.417467.
- Anchor: residual period of 529 years, zero at every great-chart boundary, all 23 residue classes at chart starts, J(n + 529) − J(n) = −22 q.
- Quantum irrelevance over 966 years: largest residual 0.499 q, window 0.996 q, mean +0.0009 q, RMS 0.284 q.
- Distributed count: the complete years 2029 and 2994 with cycles of 1 519 or 1 520 q, extra quanta equal to the annual jump, two adjacent spacings and all boundaries within q/2; about 40 000 time points mapped to coordinates and back; −1 before every zero.
- Derivation: b and g are reproduced, and the Sun is at 0° system longitude at 00.

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- Rotation axis: grid and residual, four consistent house systems, at most 3.1″ from the Swiss Ephemeris over 2029–2264.
- Lunar axis: joe of 29 or 30 q, new moons within q/2 of ν (window 0.9998 q), lunations of 34 427 469 + 0 … 4 q, about 20 000 time points mapped to coordinates and back, new moon at phase 0 and full moon at 582 912.
- Lunar derivation: Φ_0 = 65 331, A and G are reproduced; nature minus definition over 11 948 new moons has mean −0.30 min and range ±14.1 h; every system year has 12 or 13 new moons.
- Jules: 1 Ju = 5 832 409/6 250 q, system time − TT = −25 ms.
- Polar axis: 1 Ir = 1″/299 = 3.3445 mas, integer fields, field probabilities summing to 1; IERS 20 C04 with 23 648 days without gaps; one-day forecast error 0.25 Ir, growing with the horizon.
- ΔT axis: 19 996 days since 1972 without jumps, support values 1980–2020 matched to 10 ms, fields in q, one-day forecast error 0.27 ms.
- Epicycle axes: 8 solar and 8 lunar circles counted as integers; Sun 6.59 → 0.71 min, Moon 7.5 h → 3.2 min, equal to the floating-point decomposition.
- Tides and eclipses: S2 = 582 912 q, spring–neap = 582 912 joe, M2 = 12.42060 h; the nodal axis separates eclipse from no eclipse, and lunation 0 begins with the solar eclipse of 11 July 2029.
- Plato's question: three known circles recovered from a synthetic signal (periods to 10^{-6}, amplitudes to 10^{-4}, residual below 10^{-5} of the start), the same circles fitted to one half and continued over the other (residual below 10^{-4}), a Kepler ellipse fitted to 100 years and continued over 200 (below 10^{-7} AU), Kepler's equation against its series to e^4, the circle for e = 0, and for the Sun over 30 years the anomalistic year and its half as the first two circles.

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```
from zyklus23 import (Struktur, DefinierterAnker, VerteilterKalender,
Mondachse, pol_feld, platon_stand,
                      platon_vorhersage)

s = Struktur()
kal = VerteilterKalender(s, DefinierterAnker(s))   # time in q since jj, 966
years
kal.koordinaten(0).text()      # 'Karte 0, Jahr 0, Monat 0, Tag 0, Zyklus 0,
Tick 0, q 0'
kal.sprung(0)                  # 1331 extra quanta in the first year
ma = Mondachse(s)              # lunar axis, nu(x) = (9634217446503 x + 32
x^2) / 23^4
ma.koordinate(0).text()        # 'Lunation 0, Mondzyklus 43, Mondtick 2, joe
11, q 0'
s.jules_q                      # Fraction(5832409, 6250) q = 1 Jules
pol_feld(0.177494, 0.326038)   # (53, 97): pole field in Iris on 29 Sep 2026
platon_stand()["Mars"]["aequivalent"]   # 140 circles match two Kepler
ellipses (about 1.5 min)
v = platon_vorhersage("Mars")          # fitted to 100 years, measured on
the next 100
v["kepler"]["innen"], v["kepler"]["aussen"]   # (34.63, 39.16)
```

15 Discussion

Define, then measure. The construction follows the practice of modern time scales: the SI second, Terrestrial Time and the Earth rotation angle are definitions, and nature's deviation is measured and published separately (ΔT, UT1 − UTC). Here the anchor and the defined ΔT play that role. Everything inside the definition is exact and quantum-irrelevant; everything outside it is an observable series.

Analogues in machine learning. Carrying the rounding residual is error feedback in gradient compression (Karimireddy et al. 2019) and compensated summation in low-precision optimisers. As the comparison in Section 6 shows, it keeps the cumulative error bounded, whereas stochastic rounding is unbiased but grows as $\sqrt{N}$. Event-triggered spline knots correspond to lazily aggregated gradients (Chen et al. 2018), and aliasing by annual sampling to the aliasing that anti-aliased downsampling removes in CNNs (Zhang 2019). The calibration at 00 mirrors post-training quantisation: scales are fitted on calibration data and then frozen. The forecast test of Section 12 is a temporal train–test split: the training error keeps falling with the number of circles while the forecast error stalls, early stopping on the forecast error picks the number of circles, and a model class with the right inductive bias, the ellipse, generalises beyond the training window.

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Limitations. λ_0, the anchor and ΔT_def are provisional (VSOP87 and the current ΔT extrapolation), as is the lunar axis (Meeus 49.1), and will be fixed at 00 with DE440 and IERS data. The anchor is derived for 966 years and can be redefined at any great-chart boundary, where the residual is exactly zero. The system day is not a solar day and drifts against civil time by one day in 4.14 years. Polar motion is not modelled, and the lunar axis counts the mean lunation, from which true new moons deviate by up to 14.1 h. The pole forecast is deliberately simple; forecasts that use atmospheric and oceanic data are more accurate. Between 15° and 20° node distance the true orbit, not the nodal axis, decides on an eclipse, and the tidal phases refer to Greenwich and the equilibrium tide. The forecast test of Section 12 covers one split of 100 + 100 years; other windows and longer horizons may rank the models differently.

Open points. The final calibration at 00; whether to redefine the anchor after the first great chart in 2558; the format in which the series nature minus definition are published; a long-term precession model for the rotation axis beyond 2300.

Author contributions and AI disclosure

J. C. Anders conceived and named the Uniquantsystem and made all design decisions: the 23-year cycle on the 1/23 scale, the zero crossings, the −1 before every zero, quantum irrelevance, the calibration point 00 and its name jj, the units Jules, joe and Iris, and the defined anchor. The mathematical elaboration, the reference implementation, the numerical verification and the drafting of this manuscript were carried out with the support of Claude, an AI model developed by Anthropic.

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- PyEphem 4.2.1 (VSOP87): true solar crossings of λ_0 2029–2995, true equinoxes 2027–2992 and true new moons 2029–2995; positions of the Sun, Moon and Mars from jj over 100 and 200 years (Section 12)
- Observatoire de Paris: vernal-equinox year, 365.24236460 d at J2000
- Astronomical bases of calendars: Laskar polynomial of the mean tropical year
- IAU 2006 Resolution B3: $L_B = 1.550519768 \times 10^{-8}$
- ERFA era00, after Capitaine, Guinot and McCarthy (2000), A&A 355, 398–405, and the IERS Conventions (2003): Earth rotation angle
- ΔT (timekeeping): ΔT = TT − UT1; 63.83 s (2000), 69.361 s (2020)
- IERS Bulletin C 68: TAI − UTC = 37 s since 1 January 2017

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